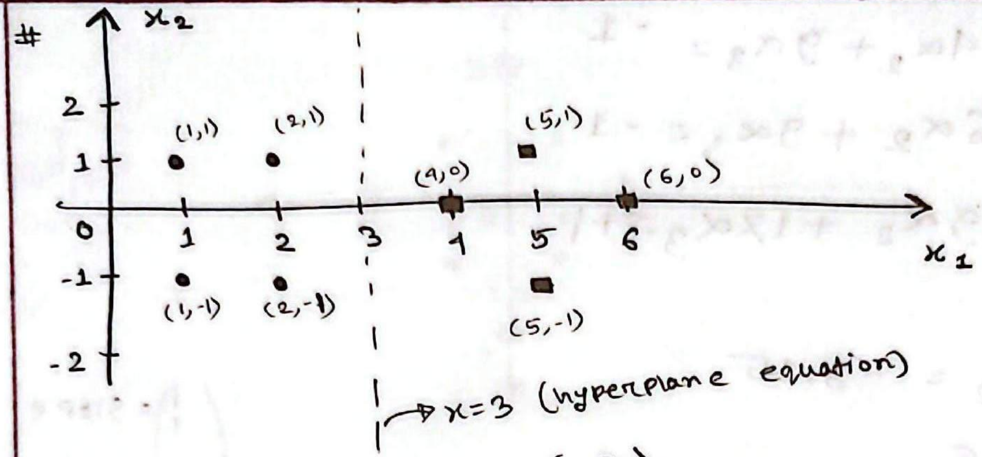
[SVM]

#



$\alpha_1 \quad S_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \qquad \tilde{S}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

$\alpha_2 \quad S_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \qquad \tilde{S}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

$\alpha_3 \quad S_3 = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \qquad \tilde{S}_3 = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_1 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_1 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_1 = -1$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_2 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_2 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_2 = -1$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_3 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_3 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_3 = +1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = +1$$

$6\alpha_1 + 4\alpha_2 + 9\alpha_3 = -1$

$4\alpha_1 + 6\alpha_2 + 9\alpha_3 = -1$

$9\alpha_1 + 9\alpha_2 + 17\alpha_3 = +1$

$\therefore \alpha_1 = \alpha_2 = -3.25$

$\alpha_3 = 3.5$

$\begin{pmatrix} \\ 1 \end{pmatrix}$ = slope

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ = vertical

$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ = Horizontal

$\tilde{w} = \sum_i \alpha_i \tilde{S}_i$

$= \alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$

$= (-3.25)\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + (-3.25)\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + 3.5 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$

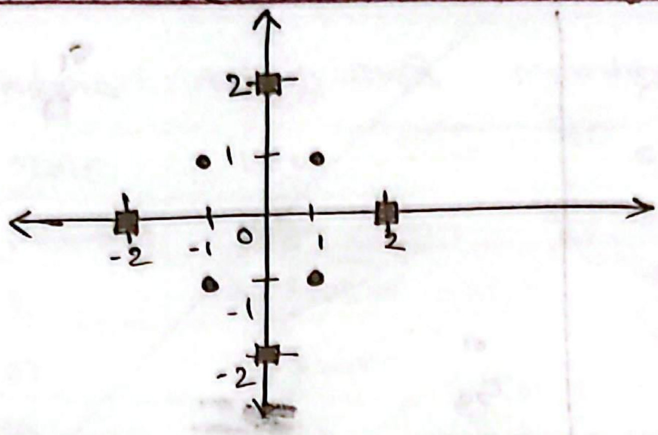
$y = w^T x + b \quad \Big| \quad w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ & $b = -3$

$\Rightarrow \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + (-3) = 0$

$\Rightarrow x_0 - 3 = 0$

$\therefore x_0 = 3$

(Kernel Trick)



Blue: $\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}-1\\1\end{pmatrix}, \begin{pmatrix}-1\\-1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}$ $\to x_1$, $\to x_2$

Red: $\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\2\end{pmatrix}, \begin{pmatrix}-2\\0\end{pmatrix}, \begin{pmatrix}0\\-2\end{pmatrix}$

$$\phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{cases} \begin{pmatrix}6 - x_1 + (x_1 - x_2)^2\\ 6 - x_2 + (x_1 - x_2)^2\end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} \geq 2 \\ \begin{pmatrix}x_1\\x_2\end{pmatrix} & \text{otherwise}\end{cases}$$

$\downarrow$

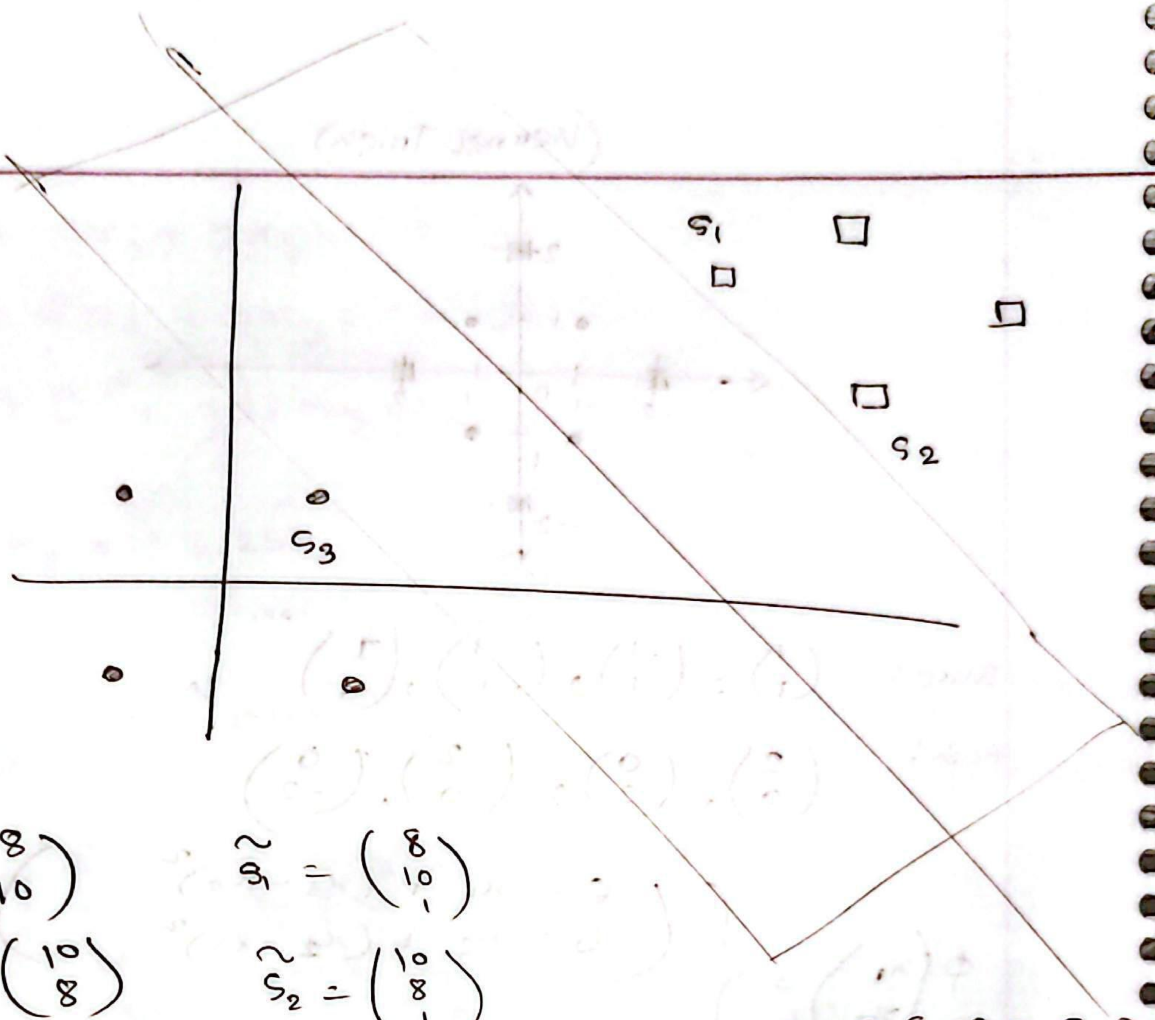
Blue: $\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}-1\\1\end{pmatrix}, \begin{pmatrix}-1\\-1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}$

$$\phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \phi\begin{pmatrix}2\\0\end{pmatrix} = \begin{pmatrix}6-2+(2-0)^2\\6-0+(2-0)^2\end{pmatrix} = \begin{pmatrix}8\\10\end{pmatrix}$$

$$\phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \phi\begin{pmatrix}0\\2\end{pmatrix} = \begin{pmatrix}6-0+(0-2)^2\\6-2+(0-2)^2\end{pmatrix} = \begin{pmatrix}10\\8\end{pmatrix}$$

$$\phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \phi\begin{pmatrix}-2\\0\end{pmatrix} = \begin{pmatrix}6+2+(-2-0)^2\\6-0+(-2-0)^2\end{pmatrix} = \begin{pmatrix}12\\10\end{pmatrix}$$

$$\phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \phi\begin{pmatrix}0\\-2\end{pmatrix} = \begin{pmatrix}6-0+(0+2)^2\\6+2+(0+2)^2\end{pmatrix} = \begin{pmatrix}10\\12\end{pmatrix}$$



α_1 $\quad s_1 = \begin{pmatrix} 8 \\ 10 \end{pmatrix} \quad \tilde{s}_1 = \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix}$

α_2 $\quad s_2 = \begin{pmatrix} 10 \\ 8 \end{pmatrix} \quad \tilde{s}_2 = \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix}$

α_3 $\quad s_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \tilde{s}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$\alpha_1 \downarrow \quad s_1 s_1 \downarrow s_2, s_3 \quad \alpha s_2 s_1 \downarrow s_2, s_3 \quad s_3 s_1 \downarrow s_2, s_3$

$$\alpha_1 \tilde{s}_1 \tilde{s}_1 + \alpha_2 \tilde{s}_2 \tilde{s}_1 + \alpha_3 \tilde{s}_3 \tilde{s}_1 = -1$$

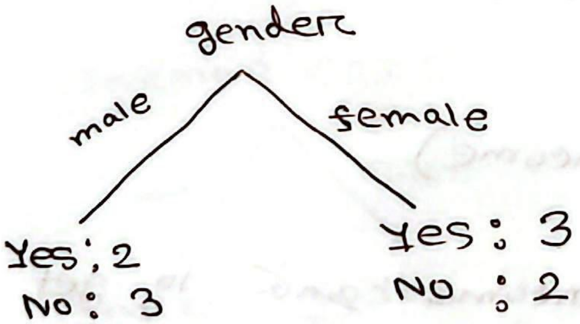
$$\alpha_1 \tilde{s}_1 \tilde{s}_2 + \alpha_2 \tilde{s}_2 \tilde{s}_2 + \alpha_3 \tilde{s}_3 \tilde{s}_2 = +1$$

$$\alpha_1 \tilde{s}_1 \tilde{s}_3 + \alpha_3 \tilde{s}_2 \tilde{s}_3 + \alpha_3 \tilde{s}_3 \tilde{s}_3 = +1$$

[CART] (classification and Regression Trees)

# ID	Gender	ActivityLevel	MonthlyIncome	BuysShoe
1	Male	Low	25	NO
2	Female	High	55	Yes
3	F	medium	48	Yes
4	m	medium	35	NO
5	m	High	60	Yes
6	F	Low	22	NO
7	F	High	70	Yes
8	m	medium	40	NO
9	F	Low	30	NO
10	m	High	65	Yes

GI (Gender)

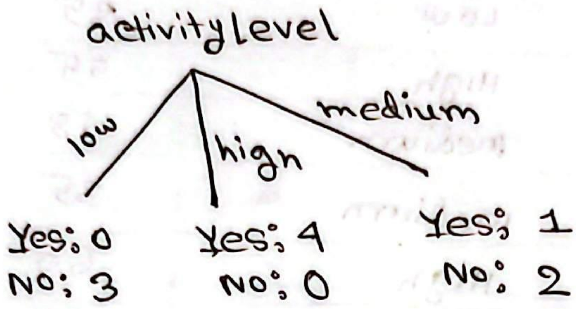


$$GI\,(gender) = \left[1 - \left(\frac{2}{2+3}\right)^2 - \left(\frac{3}{3+2}\right)^2\right] \times \frac{5}{10}$$

$$+ \left[1 - \left(\frac{3}{3+2}\right)^2 - \left(\frac{2}{3+2}\right)^2\right] \times \frac{5}{10}$$

$$= 0.24 + 0.24 = 0.48$$

GI (activity)



$$GI(\text{ActivityLevel}) = \left[1-\left(\frac{0}{0+3}\right)^2-\left(\frac{3}{0+3}\right)^2\right]\times\frac{3}{10}$$
$$+\left[1-\left(\frac{4}{4+0}\right)^2-\left(\frac{0}{0+4}\right)^2\right]\times\frac{4}{10}$$
$$+\left[1-\left(\frac{1}{1+2}\right)^2-\left(\frac{2}{1+2}\right)^2\right]\times\frac{3}{10}$$

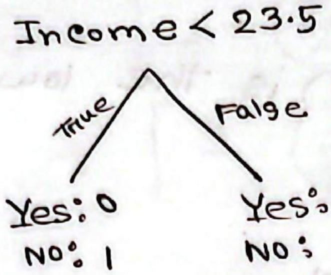
$$= 0 + 0 + 0.13 = 0.13$$

GI (monthly Income)

Sort monthly income ~~and~~ to get midpoints.

Sort (monthly Income) = 22 25 30 35 40 48 55 60 65 70

23.5 27.5 32.5 37.5 44 51.5 57.5 62.5 67.5

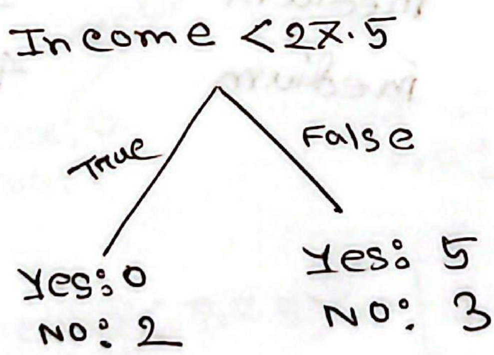


$$GI\ (\text{income} < 23.5) = \left[1 - \left(\tfrac{0}{1}\right)^2 - \left(\tfrac{1}{1}\right)^2\right] \times \frac{1}{10} + \left[1 - \left(\tfrac{5}{9}\right)^2 - \left(\tfrac{4}{9}\right)^2\right] \times \frac{9}{10}$$

$$= 0.44$$

$$GI\ (\text{income} < 27.5) = \left[1 - \left(\tfrac{0}{2}\right)^2 - \left(\tfrac{2}{2}\right)^2\right] \times \frac{2}{10} + \left[1 - \left(\tfrac{5}{8}\right)^2 - \left(\tfrac{3}{8}\right)^2\right] \times \frac{8}{10}$$

$$= 0.375$$



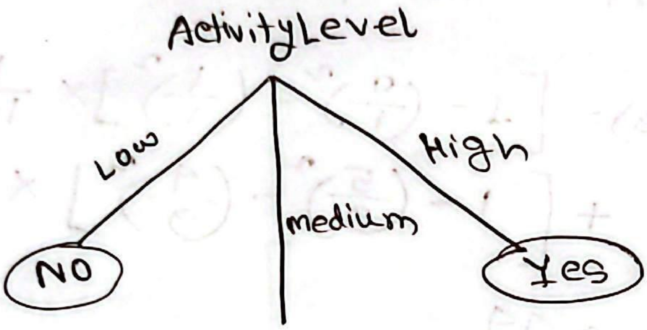
$$GI\ (\text{incom} < 32.5) = 0.285$$

⋮

we have to do it ^for^ all the midpoints

From all of the GI,

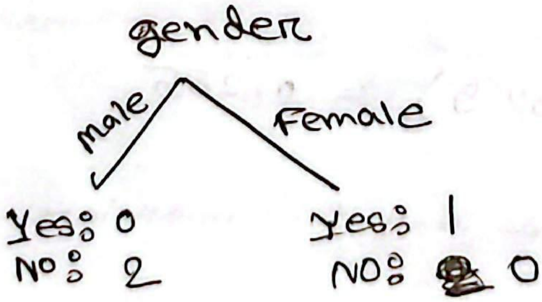
GI(activity) is the lowest



New Set

ID	Gender	ActivityLevel	MonthlyIncome	BuyShoe
3	Female	medium	48	Yes
4	male	medium	35	NO
8	male	medium	40	NO

GI (gender)



$$GI(gender) = \left[1 - \left(\frac{0}{2}\right)^2 - \left(\frac{2}{2}\right)^2\right] \times \frac{2}{3}$$

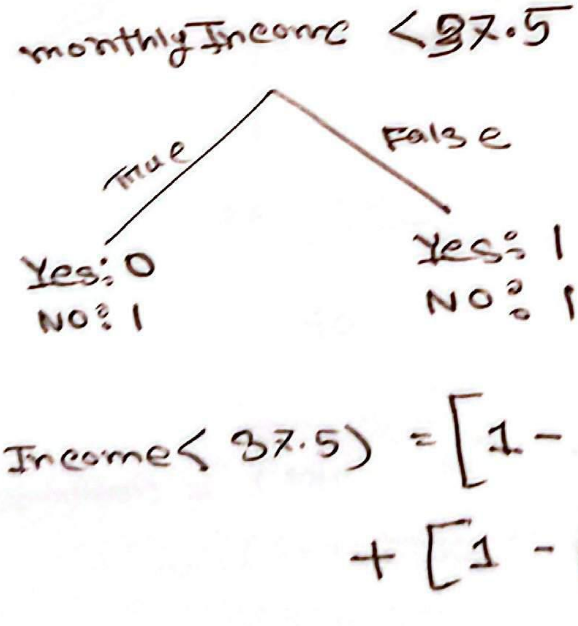
$$+ \left[1 - \left(\frac{1}{1}\right)^2 - \left(\frac{0}{1}\right)^2\right] \times \cancel{\frac{2}{3}} \; \frac{1}{3}$$

$$=$$

GI (monthlyIncome)

sorted = 35 40 48

37.5 44

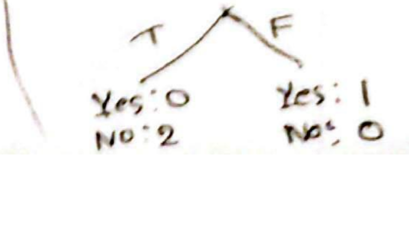


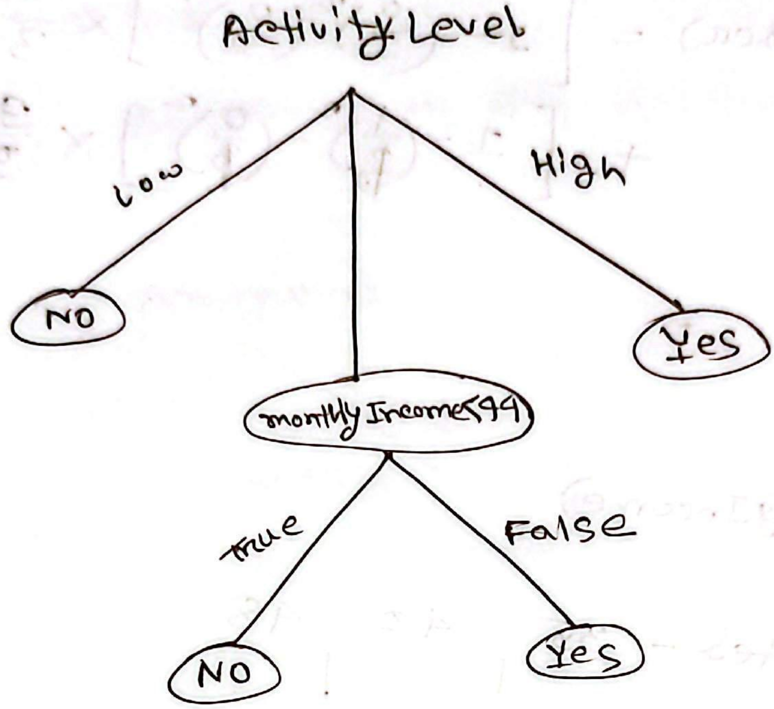
$$GI(monthlyIncome < 37.5) = \left[1 - \left(\frac{0}{1}\right)^2 - \left(\frac{1}{1}\right)^2\right] \times \frac{1}{3}$$

$$+ \left[1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2\right] \times \frac{2}{3}$$

$$= 0.33$$

$$GI(monthlyIncom < 44) = 0$$

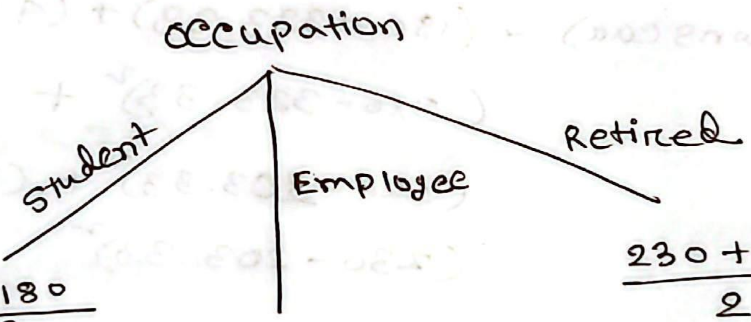



Activity Level
Low
High
No
Yes
monthly Income<49
True
False
No
Yes

Regression Task:

ID	Occupation	OwnsCar	Age	MonthlySpending
1	Student	NO	19	120
2	Student	Yes	22	180
3	Employee	NO	28	260
4	Employee	Yes	32	420
5	Retired	NO	60	230
6	Retired	Yes	70	370

SSR (occupation):



$\frac{120+180}{2} = 150$

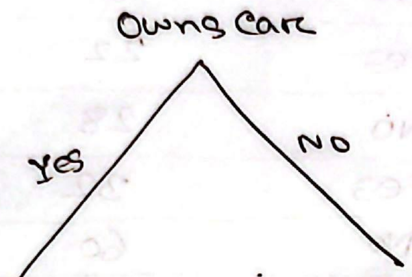
$\frac{260+420}{2} = 340$

$\frac{230+370}{2} = 300$

$$SSR(occupation) = (120-150)^2 + (180-150)^2 + (260-340)^2 + (420-340)^2 + (230-300)^2 + (370-300)^2$$

$$= 24400$$

SSR(Owns car):



$$\frac{180+420+370}{3} \qquad\qquad \frac{120+260+230}{3}$$

$$= 323.33 \qquad\qquad = 203.33$$

$$SSR(\text{Owns car}) = (180-323.33)^2 + (420-323.33)^2 +$$
$$(370-323.33)^2 +$$
$$(120-203.33)^2 + (260-203.33)^2 +$$
$$(230-203.33)^2$$

$$= 42964124$$

SSR(Age):

Sorted(19 22 28 32 60 70)

age < 22

True: $\frac{120}{1}$

False: $\frac{180 + 260 + 420 + 230 + 370}{5}$

$= 292$

$$SSR(age < 22) = (120 - 120)^2 + (180 - 292)^2 + (260 - 292)^2 + (420 - 292)^2 + (230 - 292)^2 + (370 - 292)^2$$

$=$

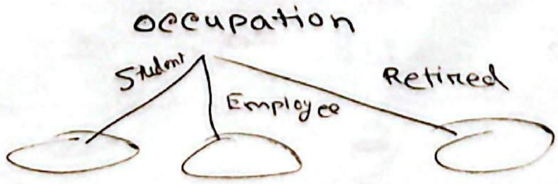
SSR (age < 28) =

SSR (age < 32) =

SSR (age < 60) =

SSR (age < 70) =

we need to find the lowest SSR



Regression Tree:

#

ID	Age	Living	IQ
1	8	village	1
2	10	city	3
3	12	village	1
4	30	village	4
5	35	city	5
6	60	city	2

Solution:

Age (Avg.)

8, 10 > 9

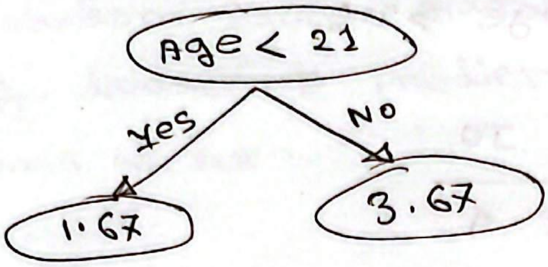
10, 12 > 11

12, 30 > 21

30, 35 > 32.5

35, 60 > 47.5

Splitter	left avg	right avg	SSR
9	$\frac{1}{1} = 1$	$\frac{3+1+4+5+2}{5} = 3$	$(1-1)^2 + (3-3)^2 + (1-3)^2 + (4-3)^2 + (5-3)^2 + (2-3)^2 = 10$
11	$\frac{1+3}{2} = 2$	$\frac{1+4+5+2}{4} = 3$	12
21	1.67	3.67	7.33 ← minimum
32.5	2.25	3.5	11.25
47.5	2.8	2	12.8
~~60~~			
City	$(5+2+3)/3 = 5.33$	$(1+1+4)/3 = 2$	$(5-5.33)^2 + (2-(5.3)^2 + (3-5.3)^2 + (1-2)^2 + (1-2)^2 + (4-2)^2$



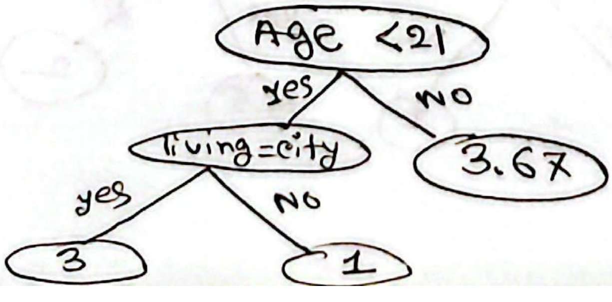
Subsetting on age < 21 :

ID	Age	Living	IQ
1	8	village	1
2	10	city	3
3	12	village	1

Age (avg) :

8 > 9
10 > 11
12

Splitter	left avg	right avg.	SSR
9	$1/1 = 1$	$\frac{3+1}{2} = 2$	$(1-1)^2+(3-2)^2+(1-2)^2 = 2$
11	$\frac{3+1}{2} = 2$	$\frac{1}{1} = 1$	2
city	$\frac{3}{1} = 3$	$\frac{1+1}{2} = 1$	0 ← minimu



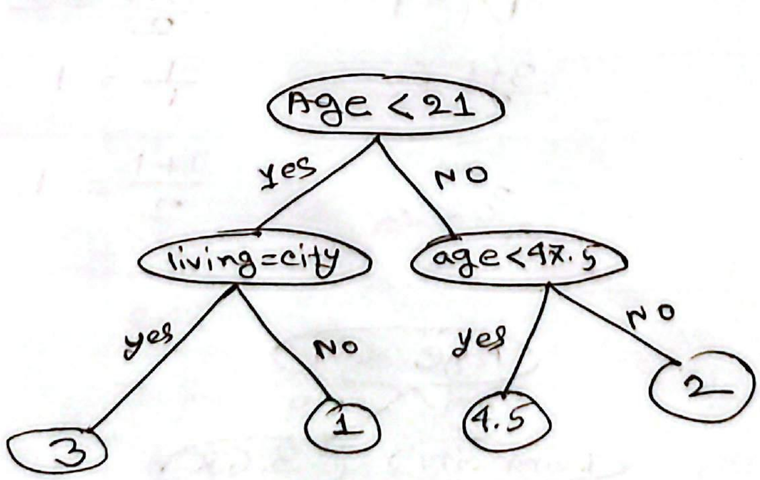
Subsetting on age >21:

ID	Age	Living	IQ
4	30	village	4
5	35	city	5
6	60	city	2

age(avg).

30, 35 > 32.5
35, 60 > 47.5

Splitter	left_avg	right_avg	SSR
32.5	$\frac{4}{1}=4$	$\frac{5+2}{2}=3.5$	4.5
✓ 47.5	4.5	2	0.5 ← minimum
city	3.5	4	4.5



[Random Forest]

Create a random forest classifier using the following dataset to predict whether a girl should play tennis or not.

Day	Outlook	Temp	Hum	wind	play Tennis
1	Sunny	Hot	High	weak	NO
2	Sunny	Hot	High	strong	NO
3	Overcast	Hot	High	weak	Yes
4	Rain	mild	High	weak	Yes
5	Rain	cool	Normal	weak	Yes
6	Rain	cool	Normal	strong	Yes
7	overcast	cool	Normal	strong	Yes
8	sunny	mild	High	weak	No
9	sunny	cool	Normal	weak	Yes
10	Rain	mild	Normal	weak	Yes
11	Sunny	mild	Normal	strong	Yes
12	Overcast	~~Hot~~ mild	High	Strong	Yes
13	overcast	~~Mild~~ Hot	Normal	weak	Yes
14	Rain	Mild	High	Strong	NO

Step 1: Bootstrapped sampling #1

Day	Outlook	Temp	Hum	Wind	Playtennis
10	Rain	Mild	Normal	weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	strong	Yes
13	Overcast	Hot	Normal	weak	Yes
14	Rain	Mild	High	Strong	NO
2	Sunny	Hot	High	Strong	NO

Step 2: Randomly take subset of features and calculate gini impurity.

feature set = {Outlook, Temp, Hum, wind}

chosen_subset = {Temp, Hum}

$$GINI\ (temp) = \left[1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2\right] \times \frac{4}{6}$$

$$+ \left[1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2\right] \times \frac{2}{6}$$

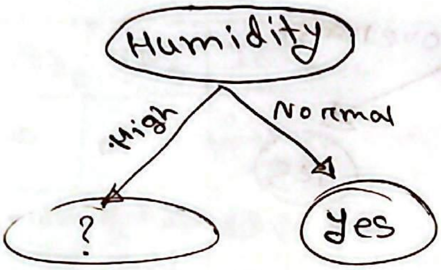
$$= 0.417$$

$$\text{GINI (Humidity)} = \left[1 - \left(\frac{3}{3}\right)^2 - \left(\frac{0}{3}\right)^2\right] \times \frac{3}{6}$$

$$+ \left[1 - \left(\frac{1}{3}\right)^2 - \left(\frac{2}{3}\right)^2\right] \times \frac{3}{6}$$

$$= 0.22123$$

Since, GINI (Temperature) > GINI (Humidity)



Randomly pick subset of features

we pick [temp, outlook]

Day	outlook	Temp	Hum	wind	Play tennis
12	overcast	mild	High	strong	Yes
14	Rain	mild	High	"	NO
2	sunny	Hot	High	"	NO

$$\text{GINI (temp)} = \left[1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2\right] \times \frac{2}{3} + \left[1 - \left(\frac{0}{1}\right)^2 - \left(\frac{1}{1}\right)^2\right] \times \frac{1}{3}$$

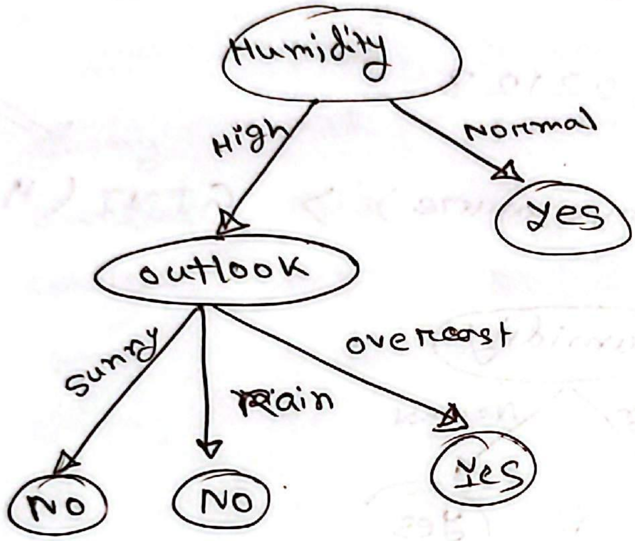
$$= 0.33$$

$$\text{GINI (outlook)} = \left[1 - \left(\frac{1}{1}\right)^2 - \left(\frac{0}{1}\right)^2\right] \times \frac{1}{3} + [1-1] \times \frac{1}{3} + [1-1] \times \frac{1}{3}$$

$$= 0$$

GINI (outlook) < GINI (temp)

so,



same thing again & again
from the start

[TF-IDF]

D1 : He is a lazy boy. She is also lazy.

D2 : Nisha is a smart girl. She is also smart.

Q : Nisha is a lazy girl.

vocabulary = { He, is, a, lazy, boy, She, also, Nisha, smart, girl, }

Term	TF_{D1}	TF_{D2}	TF_Q	Idf	$TF\text{-}IDF_{D1}$	$TF\text{-}IDF_{D2}$	$TF\text{-}IDF_Q$	$\cos(D_1, Q)$	$\cos(D_2, Q)$
He	$\frac{1}{8} = 0.125$	$\frac{0}{8} = 0$	0	$\log(\frac{2}{1}) = 0.30$	0.0375	0	0	$\frac{(0.07 \times 0.06 + 0)}{(\sqrt{0.0375^2 + 0.07^2 + 0.0375^2})(\sqrt{0.06^2 + 0.06^2 + 0.06^2})} = 0.46$	$\frac{(0.0375 \times 0.06 + (0.0375 \times 0.06))}{(\sqrt{0.0375^2 + 0.075^2 + 0.0375^2})(\sqrt{0.06^2 + 0.06^2 + 0.06^2})} = 0.471$
is	$\frac{2}{8} = 0.25$	$\frac{2}{8} = 0.25$	$\frac{1}{5} = 0.20$	0	0	0	0		
a	$\frac{1}{8} = 0.125$	$\frac{1}{8} = 0.125$	$\frac{1}{5} = 0.20$	0	0	0	0		
lazy	$\frac{2}{8} = 0.25$	$\frac{0}{8} = 0$	$\frac{1}{5} = 0.20$	$\log(\frac{2}{1}) = 0.30$	0.~~0375~~	0	0.06		
boy	$\frac{1}{8} = 0.125$	$\frac{0}{8} = 0$	0	0.30	0.0375	0	0		
She	$\frac{1}{8} = 0.125$	$\frac{1}{8} = 0.125$	0	0	0	0	0		
also	$\frac{1}{8} = 0.125$	$\frac{1}{8} = 0.125$	0	0	0	0	0		
Nisha	$\frac{0}{8} = 0$	$\frac{1}{8} = 0.125$	$\frac{1}{5} = 0.20$	0.30	0	0.0375	0.06		
smart	$\frac{0}{8} = 0$	$\frac{2}{8} = 0.25$	0	0.30	0	0.0375	0		
girl	$\frac{0}{8} = 0$	$\frac{1}{8} = 0.125$	$\frac{1}{5} = 0.20$	0.30	0	0.0375	0.06		

Right-tail = greater/higher/increased/improved ($>$)

Two tail = ($\neq$) different, not equal, has changed

Left-tail = less/lower/decreased/reduced ($<$)

[Inferential Statistics]

Population σ known $\rightarrow$ Z distribution

✓ σ is given

✓ sample size large ($n \geq 30$)

Population σ unknown $\rightarrow$ t distribution

✓ σ is not known

✓ small sample ($n < 30$)

1) Define
   - H_0: Null Hypothesis
   - H_1: Alternative hypothesis

2) Z or t

3) compute value

4) compare with critical value or p value.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2}$$

$$S = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (x_i - \bar{x})^2}$$

sample mean

population mean

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

population standard deviation

sample size

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}$$

one sample

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{(S_1)^2}{n_1} + \frac{(S_2)^2}{n_2}}}$$

$$t = \frac{\bar{d}}{\frac{S_d}{\sqrt{n}}}$$

For categorical data $\longrightarrow$ Chi-Square Test

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

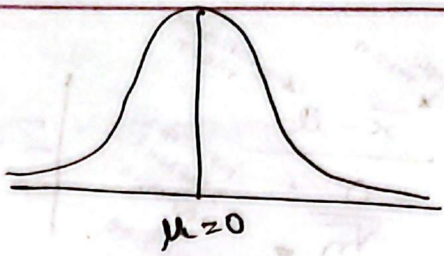
goodness of Fit test,

$$df = k - 1$$

Test of Independence,

$$df = (r-1)(c-1)$$

(15) Quiz solve:



Paired t-test

$H_0: u_d = 0$

$H_1: u_d > 0$

$n = 10$

$$t = \frac{\bar{x} - \mu}{\frac{s_d}{\sqrt{n}}} = \frac{\bar{d} - \mu}{\frac{s_d}{\sqrt{n}}}$$

$$= \frac{1.19 - 0}{\frac{0.1853}{\sqrt{10}}}$$

≈ 20.31

$$s_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{10-1}}$$

$$= \sqrt{\frac{0.309}{9}}$$

≈ 0.1853

differences are:

$d = \{1.4, 1.2, 0.9, 1.5, 1.4, 1.1, 1.1, 1.1, 1.1, 1.1\}$

$$\bar{d} = \frac{1.4 + 1.2 + 0.9 + 1.5 + 1.4 + 1.1 + 1.1 + 1.1 + 1.1 + 1.1}{10}$$

$= 1.19$

$$\sum(d_i - \bar{d})^2 = (1.4-1.19)^2 + (1.2-1.19)^2 + (0.9-1.19)^2 + (1.5-1.19)^2 + (1.4-1.19)^2 + (1.1-1.19)^2 + (1.1-1.19)^2 + (1.1-1.19)^2 + (1.1-1.19)^2 + (1.1-1.19)^2$$

$$= 0.309$$

$$t_{critical} = 1.833$$

since,

$20.31 > 1.833$; we reject H_0

Interpret : At the 5% significance level there is strong statistical evidence that the stress management training program reduced stress-related absenteeism among the employees.

The estimated avg. reduction is about 1.19 hours per week.

#

A(2,3)
B(3,4) $\longrightarrow +1$
C(4,1) $\longrightarrow -1$
D(5,2)

A(3,4) $\longrightarrow +1$
B(4,1) $\longrightarrow -1$

C(2,3)
D(5,2)

$$\alpha_A y_A + \alpha_B y_B = 0$$

$$\Rightarrow \alpha_A(+1) + \alpha_B(-1) = 0$$

$$\Rightarrow \alpha_A - \alpha_B = 0 \quad \text{——— (i)}$$

For A,

$$\alpha_A y_A (x_A \cdot x_A) + \alpha_B y_B (x_B \cdot x_A) + b = +1$$

$$\Rightarrow \alpha_A (1) \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \alpha_B (-1) \begin{bmatrix} 4 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} + b = 1$$

$$\Rightarrow \alpha_A (25) + \alpha_B (-1)(16) + b = 1$$

$$\Rightarrow 25\alpha_A - 16\alpha_B + b = 1$$

$$\Rightarrow 25\alpha_A - 16\alpha_B + b = 1 \quad \text{——— (ii)}$$

For B,

$$\alpha_A y_A (x_A \cdot x_B) + \alpha_B y_B (x_B \cdot x_B) + b = -1$$

$$\Rightarrow \alpha_A (\cancel{0}1) \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} + (-\alpha_B) \begin{bmatrix} 4 \\ 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} + b = -1$$

$$\Rightarrow \alpha_A (\cancel{0}1)(16) + \alpha_B (-17) + b = -1$$

$\Rightarrow 16\alpha_A - 17\alpha_B + b = -1$ ——— (iii)

$\alpha_A = \frac{1}{5}$

$\alpha_B = \frac{1}{5}$

$b = -\frac{4}{5}$

$w = \alpha_A y_A x_A + \alpha_B y_B x_B$

$= \frac{1}{5}(1)\begin{bmatrix} 3 \\ 4 \end{bmatrix} + \frac{1}{5}(-1)\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} + \begin{bmatrix} -\frac{4}{5} \\ -\frac{1}{5} \end{bmatrix} = \begin{bmatrix} -\frac{1}{5} \\ \frac{3}{5} \end{bmatrix}$; $b = -\frac{4}{5}$

$w^T x + b = 0$

$\Rightarrow \begin{bmatrix} -\frac{1}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \left(-\frac{4}{5}\right) = 0$

$\Rightarrow -\frac{1}{5}x + \frac{3}{5}y - \frac{4}{5} = 0$

$\Rightarrow -x + 3y - 4 = 0$

$\Rightarrow 3y = x + 4$

$\therefore y = \frac{x}{3} + \frac{4}{3}$

[Big Data]

d_1 = "Big data is powerful"
d_2 = "Big " " scalable"
d_3 = "Data analytics is powerful"

m_1 = { ("Big", 1), ("data", 1), ("is", 1), ("powerful", 1) }
m_2 = { ("Big", 1), ("data", 1), ("is", 1), ("scalable", 1) }
m_3 = { ("Data", 1), ("analytics", 1), ("is", 1), ("powerful", 1) }

g_1 = ("Big", [1, 1])
g_2 = ("Data", [1])
g_3 = ("analytics", [1])
g_4 = ("data", [1, 1])
g_5 = ("is", [1, 1, 1])
g_6 = ("powerful", [1, 1])
g_7 = ("scalable", [1])

f_{reduce} (Big, [1, 1]) ⟶ (Big, 2)

(Data [1]) ⟶ (Data, 1)

(analytics, 1)

(data, 2)

(is, 3)

(powerful, 2)

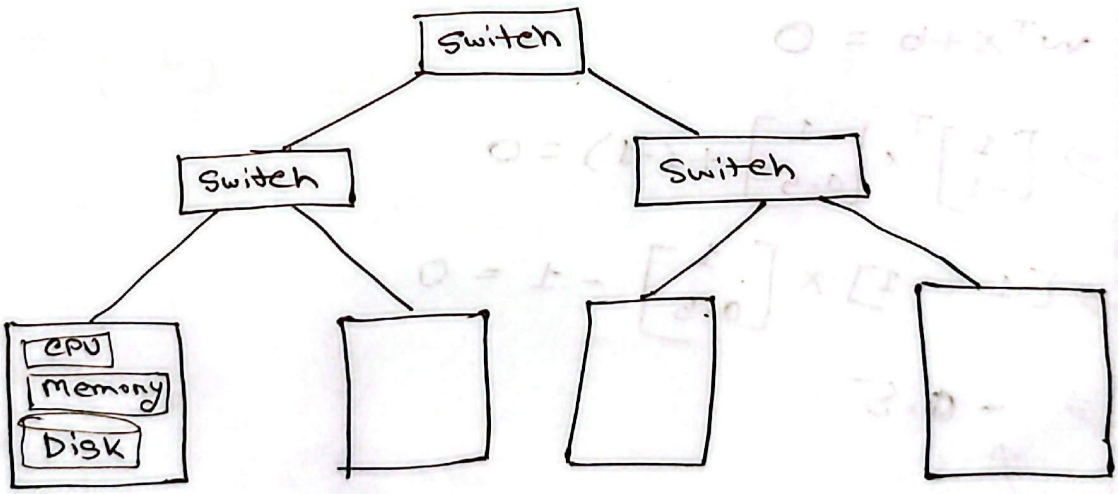
(scalable, 1)

Big Data Analytics

5V

- Volume (scale and magnitude of data)
- Velocity (speed at which data is generated, collected and processed)
- Variety (diversity of data types, formats and sources)
- Veracity (quality, reliability and trustworthiness of data)
- Value (usefulness and actionable insight)

(Cluster Architecture)



Each rack contains 16 - 64 nodes

(Distributed file system)

✓ chunk servers

✓ master Node

✓ client library for file access

map Reduce

- → MAP
- → Reduction

GRA Sir Lecture

minimize : $\frac{1}{2} |\vec{w}|^2$

Subject to the constraints, $y_i(\vec{w}\cdot\vec{x_i} + b) \geq 1$

$$\Rightarrow y_i(\vec{w}\cdot\vec{x_i} + b) - 1$$

$$\therefore L = \frac{1}{2}|w|^2 - \sum_i \alpha_i \left(y_i(\vec{w}\cdot\vec{x_i} + b) - 1\right)$$

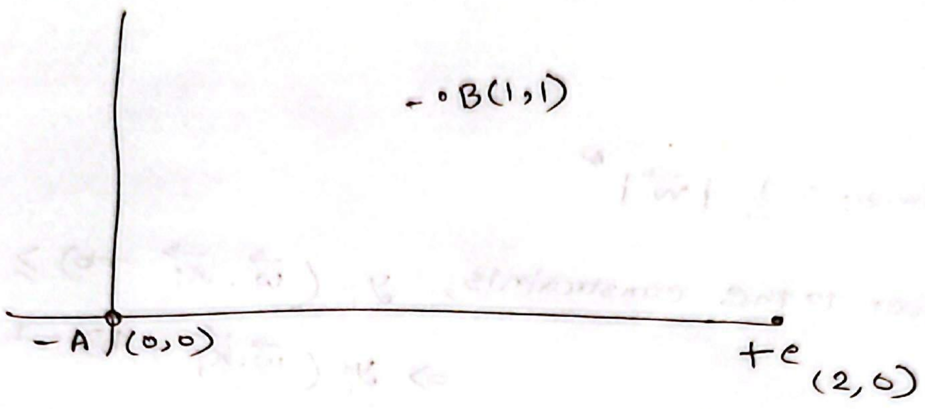
$$\frac{\partial L}{\partial b} = 0 - \sum (0 + \alpha_i y_i (1) - 0) = 0$$

$$\Rightarrow -\sum \alpha_i y_i = 0$$

$$\Rightarrow \sum \alpha_i y_i = 0 \quad \text{———} \quad (i)$$

$$\frac{\partial L}{\partial w} = \cancel{\frac{1}{2}} \times \cancel{2} w - \sum \alpha_i y_i x_i (1) + 0 = 0$$

$$\Rightarrow w = \sum_i \alpha_i y_i x_i \quad \text{———} \quad (ii)$$



$$\alpha_A y_A + \alpha_B y_B + \alpha_C y_C = 0$$

class level

$$\Rightarrow \alpha_A(-1) + \alpha_B(-1) + \alpha_C(+1) = 0$$

$$\Rightarrow -\alpha_A - \alpha_B + \alpha_C = 0 \quad \ldots\ldots\ldots\ldots (1)$$

$$\alpha_A y_A (x_A \cdot x_C) +$$
$$\alpha_B y_B (x_B \cdot x_C) +$$
$$\alpha_C y_C (x_C \cdot x_C) + b = +1$$

$$\Rightarrow \alpha_A(-1)\cdot \begin{bmatrix}0\\0\end{bmatrix}\begin{bmatrix}2\\0\end{bmatrix} + \alpha_B(-1)\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}2\\0\end{bmatrix}$$
$$+ \alpha_C(1)\begin{bmatrix}2\\0\end{bmatrix}\begin{bmatrix}2\\0\end{bmatrix} + b = 1$$

$$\Rightarrow -\alpha_A(0) - \alpha_B(2) + \alpha_C(4) + b = +1$$

$$\Rightarrow -2\alpha_B + 4\alpha_C + b = 1 \quad \ldots\ldots\ldots (2)$$

For A,

$$\alpha_A y_A (x_A \cdot x_A) + \alpha_B y_B (x_B \cdot x_A) + \alpha_C y_C (x_C \cdot x_A) + b = 1$$

$$\Rightarrow \alpha_A (-1) \begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \alpha_B (-1) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} +$$

$$\alpha_C (1) \begin{bmatrix} 2 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + b = -1$$

$$\Rightarrow -\alpha_A(0) - \alpha_B(0) + \alpha_C(0) + b = -1$$

$$\therefore b = -1 \quad \ldots\ldots\ldots (3)$$

For B,

$$\alpha_A y_A (x_A \cdot x_B) + \alpha_B y_B (x_B \cdot x_B) + \alpha_C y_C (x_C \cdot x_B) + b = -1$$

$$\Rightarrow \alpha_A (-1) \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \alpha_B (-1) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \alpha_C (+1) \begin{bmatrix} 2 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b = -1$$

$$\Rightarrow -\alpha_A(0) - \alpha_B(2) + \alpha_C(2) + b = -1$$

$$\Rightarrow -2\alpha_B + 2\alpha_C + b = -1 \quad \ldots\ldots\ldots (4)$$

After solving 4 equations,

$$\alpha_A = 0$$

$$\alpha_B = 1$$

$$\alpha_C = 1$$

$$b = -1$$

$$w = \alpha_A y_A x_A + \alpha_B y_B x_B + \alpha_C y_C x_C$$

$$= 0 + 1 \times (-1) \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \times (+1) \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$b = -1$$

$$w^T x + b = 0$$

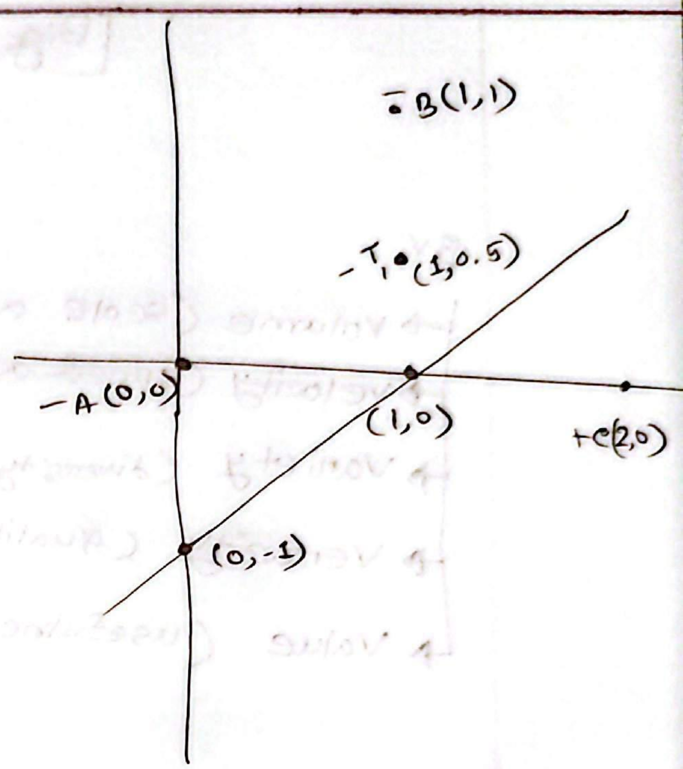
$$\Rightarrow \begin{bmatrix} 1 \\ -1 \end{bmatrix}^T + -1 = 0$$

$$\Rightarrow \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + (-1) = 0$$

$$\Rightarrow x - y - 1 = 0$$

line equation

$x - y - 1 = 0$



if $x = 0$;

$y = -1 \longrightarrow (0, -1)$

if $y = 0$;

$x = 1 \longrightarrow (1, 0)$

Test case:

$T_1 (1, 0.5)$

$w^T x + b = 0$

$\Rightarrow \begin{bmatrix} 1 \\ -1 \end{bmatrix}^T \times \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + (-1) = 0$

$\Rightarrow \begin{bmatrix} 1 & -1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} - 1 = 0$

$\Rightarrow -0.5$

negative class